

Package ‘iRamat’

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Type Package

Title Statistical analysis on Archaeomaterial

Description Statistical analysis on Archaeomaterial.

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URL <https://github.com/iramatiRamat>

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NeedsCompilation no

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 chrono

Create a timeline of a selected dataset

Description

This function parses EDTF-formatted dates from a dataset and returns one or two ggplot2 timelines: a site-based timeline and, optionally, a PeriodO timeline for comparison.

Usage

```
chrono(
  d,
  use_periodo = FALSE,
  periodo_authority = "https://n2t.net/ark:/99152/p02chr4",
  seriated = TRUE,
  all_dates = FALSE,
  time_match = 0.9,
  verbose = TRUE
)
```

Arguments

<code>d</code>	A data frame containing at least a column 'edtf' with EDTF dates and a column 'context_name'.
<code>use_periodo</code>	Logical, if TRUE (default FALSE), also fetch and plot periods from a PeriodO authority.
<code>periodo_authority</code>	A string. URL of a PeriodO authority (default: "https://n2t.net/ark:/99152/p02chr4").
<code>seriated</code>	Logical, if TRUE (default), reorder sites/periods by earliest start date.
<code>all_dates</code>	Logical, if TRUE, include sites with missing/NA dates (default: FALSE).
<code>time_match</code>	Numeric between 0 and 1. Minimum proportion of temporal overlap required between dataset range and a period to include the period (default: 0.9).
<code>verbose</code>	Logical, if TRUE (default), print progress messages.

Value

A list of ggplot2 objects with elements:

sites A timeline of dataset sites (always returned).

periodo A PeriodO timeline, if 'use_periodo = TRUE', otherwise NULL.

Examples

```
## Not run:
df <- db_api_connect()
plots <- chrono(df$dataset_adisser17, use_periodo = TRUE)
ggpubr::ggarrange(plots$sites, plots$periodo, heights = c(1,2), ncol = 1, align = "v")

## End(Not run)
```

db_api_connect	Connect the CHIPS DB API and return an R object (dataframe, etc)
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Description

Connect the CHIPS DB API and return an R object (dataframe, etc.). The default dataset is dataset_adisser17, accessed at <http://157.136.252.188:3000/dataset_adisser17>.

If all_datasets = TRUE, all datasets whose description is "Dataset" in the GitHub-hosted table https://github.com/iramat/iramat-dev/blob/main/dbs/chips/urls_data.tsv are downloaded (using their url_data column), merged into a single data.frame, and a new column dataset_name is added to indicate the source dataset for each row.

Usage

```
db_api_connect(
  d = NA,
  api_url = "http://157.136.252.188:3000/dataset_adisser17",
  api_list =
    "https://raw.githubusercontent.com/iramat/chips/hugo-files/static/data/urls_data.tsv",
  output_format = "dataframe",
  verbose = TRUE,
  all_datasets = FALSE
)
```

Arguments

d	A hash object. If none is provided, a new one will be created. Ignored when all_datasets = TRUE.
api_url	An URL landing to an API. Default: "http://157.136.252.188:3000/dataset_adisser17". Ignored when all_datasets = TRUE.
api_list	An URL landing to a TSV file listing API URLs. Default: "https://raw.githubusercontent.com/iramat/chips/hugo-files/static/data/urls_data.tsv". Ignored when all_datasets = FALSE.
output_format	The selected output format. Default "dataframe" (currently not used).
verbose	if TRUE (by default), verbose.
all_datasets	Logical. If TRUE, download all datasets with description == "Dataset" listed in the GitHub table https://github.com/iramat/iramat-dev/blob/main/dbs/chips/urls_data.tsv , merge them into a single data.frame, and add a column dataset_name with the name of the source dataset (last component of the API URL). Default FALSE.

Value

* If all_datasets = FALSE (default): a hash where each key is the dataset label (last part of api_url) and each value is a dataframe. * If all_datasets = TRUE: a single merged data.frame with an additional column dataset_name.

Examples

```
# Default behaviour: one dataset stored in a hash
df_hash <- db_api_connect()
head(df_hash$dataset_adisser17)

# All CHIPS datasets listed in urls_data.tsv, merged into a single dataframe
df_all <- db_api_connect(all_datasets = TRUE)
head(df_all)
```

periodo

Create a timeline of periods from a PeriodO authority

Description

This function reads a PeriodO authority file (CSV + JSON metadata) and returns a ggplot2 timeline of the defined periods. Optionally, the timeline can be restricted to a given temporal range, and periods can be filtered by the amount of overlap with that range.

Usage

```
periodo(
  periodo_authority = "https://n2t.net/ark:/99152/p02chr4",
  use_periodo = FALSE,
  min_date = NA,
  max_date = NA,
  time_match = NA,
  location = NA,
  seriated = TRUE,
  verbose = TRUE
)
```

Arguments

periodo_authority	A string. URL of a PeriodO authority (default: "https://n2t.net/ark:/99152/p02chr4", INRAP).
use_periodo	Logical, if TRUE, the function is used as a subroutine (typically called from [chrono()]) and the plot title is suppressed.
min_date	Optional numeric. Lower bound of the time interval (most ancient date). Default: 'NA' (no lower bound).
max_date	Optional numeric. Upper bound of the time interval (most recent date). Default: 'NA' (no upper bound).
time_match	Numeric between 0 and 1. Minimum proportion of temporal overlap required between a period and the given [min_date, max_date] interval for the period to be included. Default: 'NA' (no filtering).
location	Optional string. If provided, only periods whose 'spatial_coverage' field contains this string will be included (e.g. "France"). Default: 'NA' (no filtering).
seriated	Logical, if TRUE (default), reorder periods by earliest start date.
verbose	Logical, if TRUE (default), print progress messages.

Value

A list with one ggplot2 object:

periodo A timeline of PeriodO periods.

Examples

```
## Not run:  
# Default authority (INRAP)  
periodo()  
  
# Restrict to a specific interval and require exact overlap  
periodo(min_date = -700, max_date = 0, use_periodo = TRUE, time_match = 1)  
  
# Authority ArkeOpen, France only  
periodo(periodo_authority = "http://n2t.net/ark:/99152/p09hq4n", min_date = -500, max_date = 500, use_periodo = TRUE)  
  
## End(Not run)
```

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